05 - Interpretability

Goal: see what the models use to choose a category.

- **A. Text:** which words decide the category, for the SVM and for CamemBERT.
- **B. Fusion:** how much the final Random Forest relies on CamemBERT, the SVM and the image.

A. Text: which words decide the category?

We look at the two text models:

- **SVM:** it gives each word a weight for each category. We can read these weights directly.
- **CamemBERT:** it has no word weights. We remove one word at a time and see what changes.

A1. Train the SVM

```
Words and word pairs: 460326
Categories: 27
```

A2. The SVM's top words for each category

For each category, the 8 words that push most toward it.

Code	Category	Top words
10	Used Books	edition de, and, in, der, die, cookbook, dico, und
40	Games	magideal, import, ensemble complet, la description, pal, simulator, essentials, quickio
50	Game Accessories	gaming, ps4, pour nintendo, manette, pour, skylanders, joystick, amiibo
60	Consoles	console, nom de, sony playstation, nom, pack xbox, 500 go, console de, bundle
1140	Figurines	figurine, shirt, magnet, figurines, gundam, mug, buste, watermelon
1160	Cards	carte, mtg, vf, vo, commune, pokemon, rare, attax
1180	Wargames	heroclix, warhammer, oop, miniatures, 40k, shadowrun, wargame, elfe
1280	Toys	peluche, puzzle, hélicoptère, poupée, doudou, playmobil, lego, kinder
1281	Board Games	propos, jeu, monopoly, tarot, cards, trivial, asmodee, cartes
1300	RC Models	générique, drone, maquette, 43, ho, dji, altaya, matchbox
1301	Baby Socks	auucne, baby foot, fléchettes, billard, bce, harrows, suprême, chaussettes
1302	Outdoor Toys	stream, trampoline, pêche la, nerf, balançoire, tricycle, draisienne, pêche
1320	Baby	bébé, hochet, poussette, bavoir, langer, chicco, playshoes, biberon
1560	Furniture	kitchen supplies, sweetheart93, supplies, fournitures de, sweetheart, meuble, germenly, fournitures
1920	Textiles	cdaffaires, coussin, oreiller, taie, rideaux, manique, couette, pillow
1940	Food	bio, capsules, sucre, haribo, 100g, 250g, bonbons, goût
2060	Decor	lujuik, led, decoration de, horloge, decoration, décor germenly, housekeeping, décoration
2220	Pets	pour chien, pet, chien, pour chat, laisse, chiens, zolux, chiot
2280	Magazines	du 01, 06, 04, 08, magazine, du 15, 03, hors série
2403	Book Lots	tomes, lot, volumes, livres, revues, numéros, éditions, interview
2462	Used Games	jeux, ps2, pc, vtech, xbox, ps3, wii, jeu jaquette

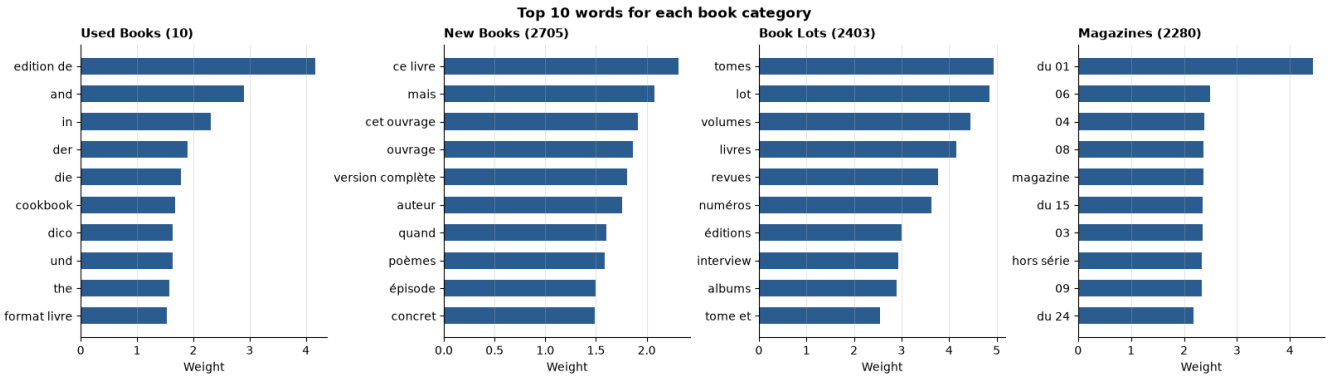
Code	Category	Top words
2522	Stationery	lindner, stylo, a4, agenda, clairefontaine, marque page, enveloppes, encre
2582	Garden Furniture	de jardin, okadaopa, de habitat, kitchen, jardin, terrasse, stainless, parasol
2583	Pool	piscine, spa, intex, sauna, piscines, pompe, cellule, bestway
2585	Garden Tools	tools, arrosage, tondeuse, outil, tronçonneuse, lame, gardena, jardinage
2705	New Books	ce livre, mais, cet ouvrage, ouvrage, version complète, auteur, quand, poèmes
2905	PC Games	en téléchargement, téléchargement, jeu en, forts de, points forts, forts, points, configuration requise

Analysis

- Most top words make sense: *piscine* (pool) for Pool, *chien* (dog) for Pets, *warhammer* for Wargames.
- Some top words are seller names, such as *sweetheart93*. The model learned who sells what, which may not work for new sellers.
- For books, the top words are about writing style: New Books uses words from a book summary (*ce livre*, *auteur*), Used Books uses foreign words (*and*, *der*) because many used books have foreign titles.

A3. Why the four book categories get mixed up

The top 10 words of each book category.



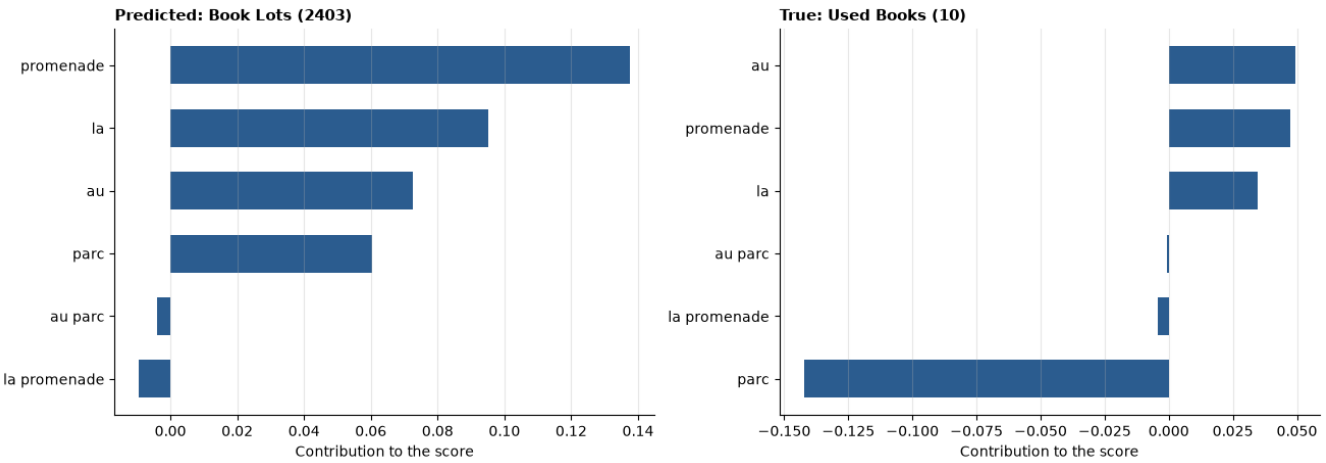
Analysis

- Each book category has a few strong words: *lot* and *tomes* for Book Lots, dates such as *du 01* for Magazines.
- A book title without one of these words gives the SVM almost nothing to go on.
- This is why book categories make 7 of the 10 most confused pairs.

A4. One SVM mistake, word by word

A used book that the SVM calls a book lot. Each bar shows how much one word pushes toward the category.

```
Title: la promenade au parc
```




```
Score for Book Lots (2403): −0.50
Score for Used Books (10):  −0.68
```

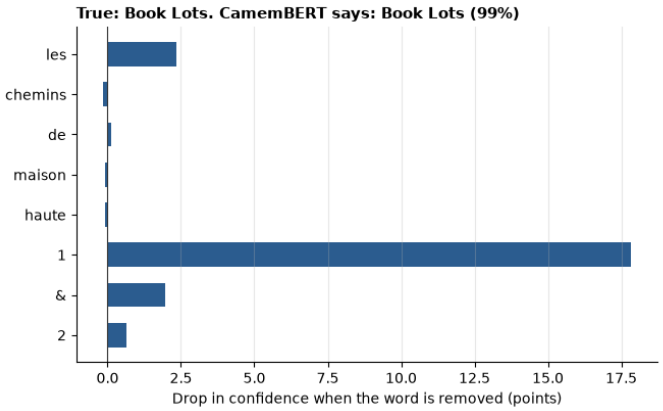
Analysis

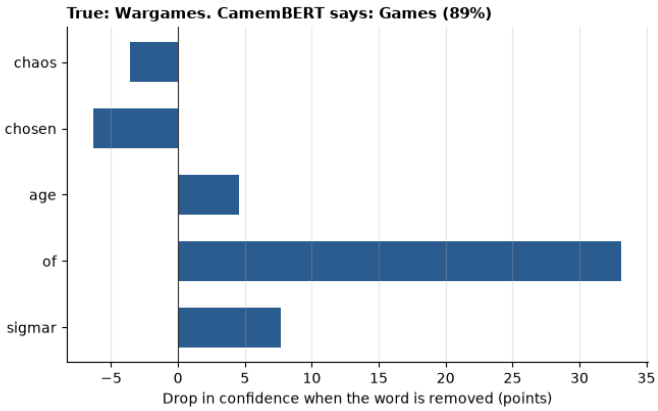
- The title is only *la promenade au parc* ("the walk in the park"), with no description.
- None of these words is a book word, so both scores are low and close. Book Lots wins by a small margin.
- When the text is short and generic, a text model has nothing to go on. The image can help here (notebook 04).

A5. Which words CamemBERT uses

CamemBERT has no word weights. So we remove one word at a time and see how much its confidence drops. A big drop means the word mattered.

Two short titles: one CamemBERT gets right, one it gets wrong.





Analysis

- **Left (right answer):** removing *1* drops the confidence by 18 points. CamemBERT reads *1 & 2* as "volumes 1 and 2", so a lot of several books. The SVM got this one wrong.
- **Right (wrong answer):** *of* matters most (33 points). CamemBERT reads *age of ...* like a video game title (*Age of Empires*) and says Games. The SVM gets it right through *sigmar*, a rare Warhammer word.
- So the two text models make different mistakes. That is why combining them helps.

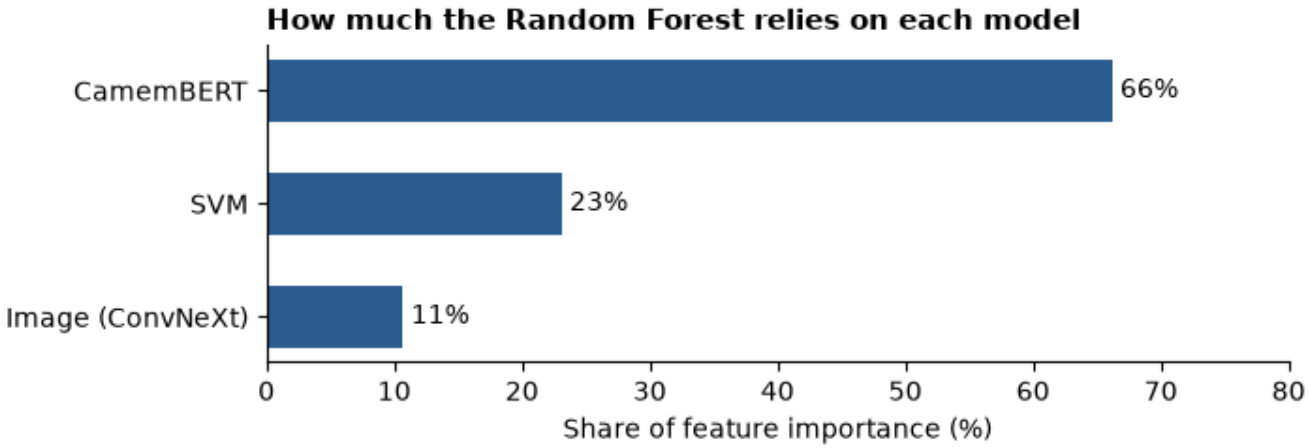
Text conclusion

- The SVM decides on single words: clear product words, but also seller names.
- CamemBERT reads words together. This helps for books, but a familiar pattern can mislead it.
- Short book titles with no clear word are hard for both.

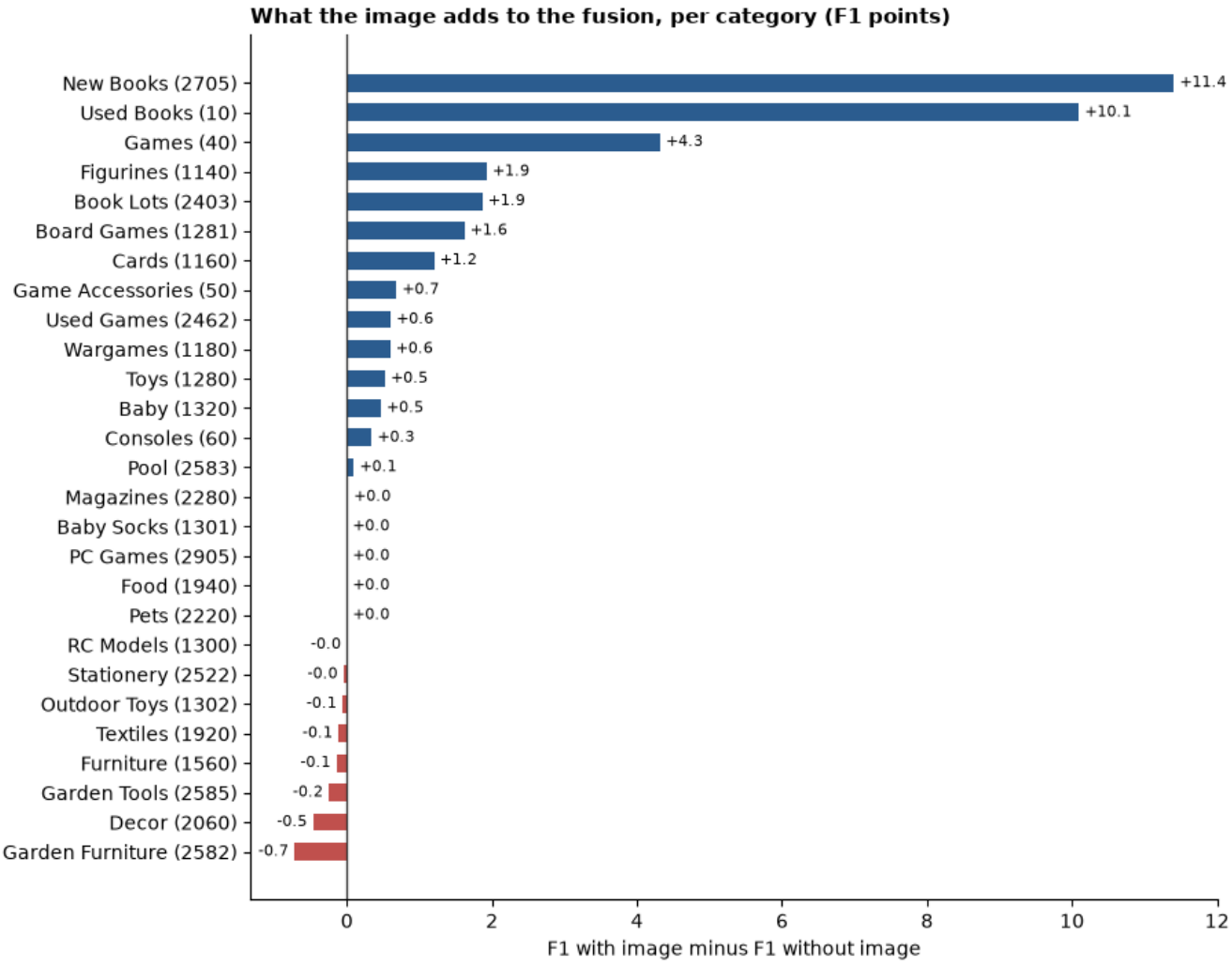
B. Fusion

The final model is a Random Forest that takes 81 inputs: 27 probabilities from CamemBERT, 27 from the SVM and 27 from the ConvNeXt image model (notebook 04). We look at it in two ways:

1. **Feature importance:** how often, and how usefully, the trees split on the inputs of each model.
2. **Image contribution per category:** the same Random Forest with and without the image, scored with the same 5-fold cross-validation as in notebook 04.




```
Weighted F1 with image:    0.8988
Weighted F1 without image: 0.8873
```



Analysis

- The Random Forest relies mostly on CamemBERT (about two thirds of the importance), then on the SVM (23%), then on the image (11%).
- Without the image, the fusion scores 0.8873. With it, 0.8988 (+1.2 points).
- The image adds almost everything on two categories: New Books (+11.4 points) and Used Books (+10.1). Text cannot tell if a book is new or used; the photo can (a clean cover image against a photo of an old copy).
- It also helps on Games (+4.3), where the box picture shows what the text does not say.
- For most other categories the image adds nothing or very little. Its small losses (at most -0.7 points) are within noise.

Conclusion

- **Text:** the SVM relies on single words, including seller names. CamemBERT reads words together, which helps for books.
- **Fusion:** CamemBERT does most of the work. The image adds little overall, but it is decisive for telling new books from used books.